

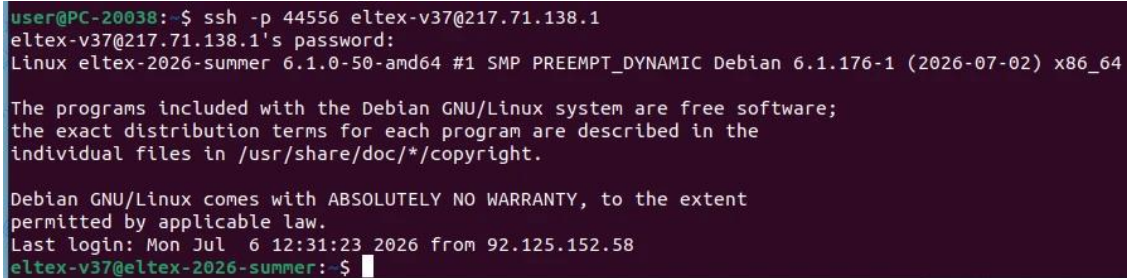
Царенкова В. А. АВТ-342

Модуль 1

Практика 1

Раздел 1

ssh -p 44556 eltex-v37@217.71.138.1

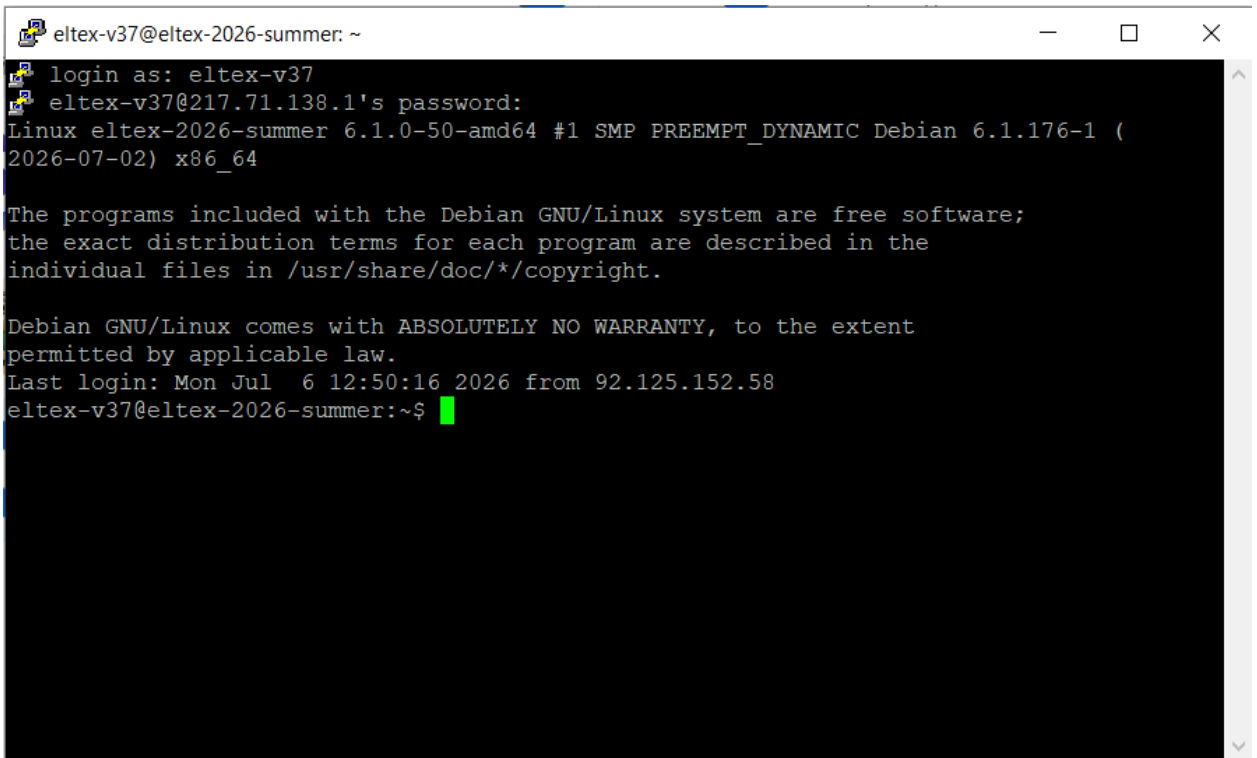


```
user@PC-20038:~$ ssh -p 44556 eltex-v37@217.71.138.1
eltex-v37@217.71.138.1's password:
Linux eltex-2026-summer 6.1.0-50-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.176-1 (2026-07-02) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Jul  6 12:31:23 2026 from 92.125.152.58
eltex-v37@eltex-2026-summer:~$
```

Рисунок 1 – Подключение ssh



```
eltex-v37@eltex-2026-summer: ~
login as: eltex-v37
eltex-v37@217.71.138.1's password:
Linux eltex-2026-summer 6.1.0-50-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.176-1 (
2026-07-02) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Jul  6 12:50:16 2026 from 92.125.152.58
eltex-v37@eltex-2026-summer:~$
```

Рисунок 2 – Подключение putty

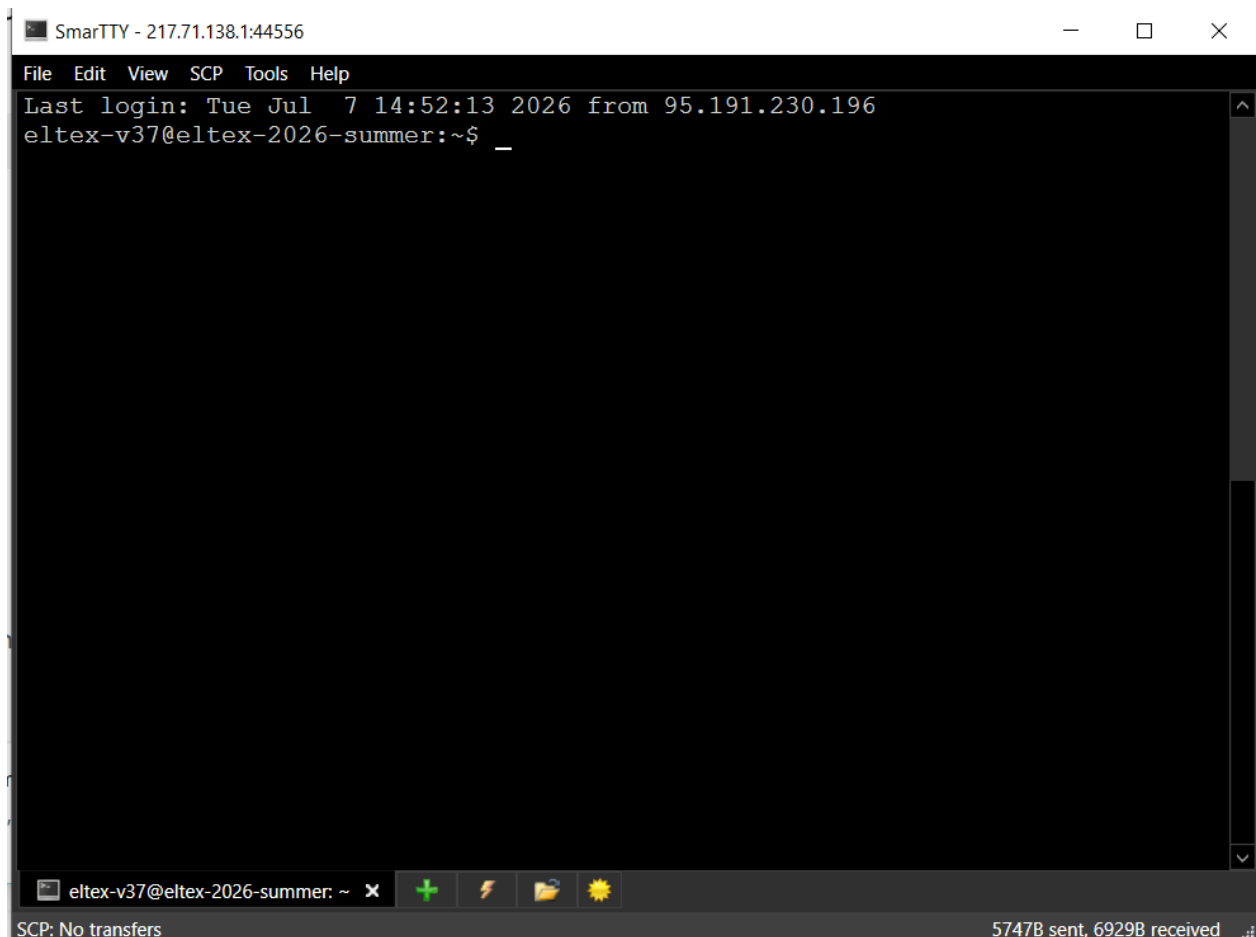


Рисунок 3 – Подключение smarty

screen

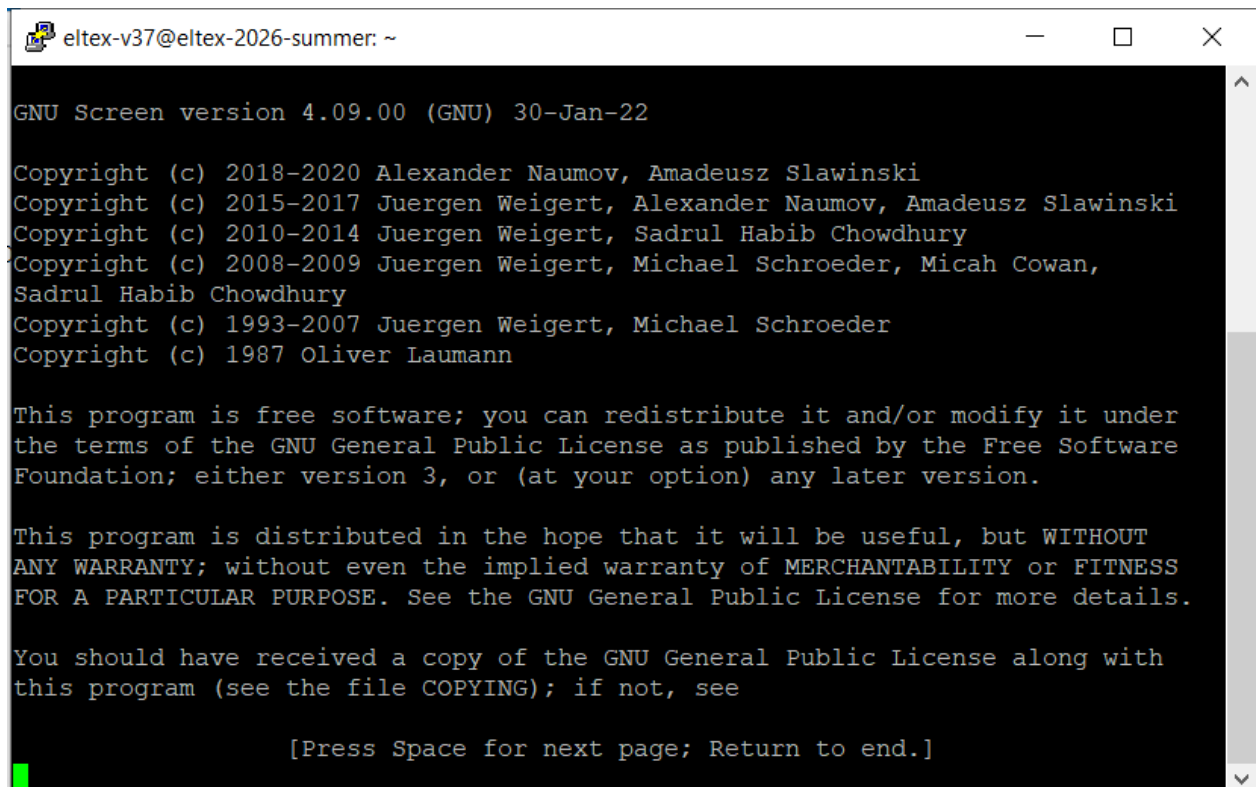


Рисунок 4 – Запуск новой сессии утилиты screen

W

```
eltex-v37@eltex-2026-summer: ~  
eltex-v37@eltex-2026-summer:~$ w  
15:13:09 up 1 day, 11:55, 16 users, load average: 0.00, 0.02, 0.00  
USER      TTY      FROM          LOGIN@      IDLE   JCPU   PCPU WHAT  
tech      pts/0    37.193.179.176 Mon03       7:42m  0.02s  0.02s -bash  
eltex-v1  pts/2    :pts/158:S.0   Mon13      25:29m  1:49   4.86s SCREEN -S top -  
eltex-v2  pts/8    212.164.65.3   15:01      3.00s  0.04s  0.04s -bash  
eltex-v1  pts/14   :pts/21:S.0    Mon13      26:01m  1:52   4.93s SCREEN -S top -  
eltex-v3  pts/20   95.191.230.196 15:11      1.00s  0.00s  ?      screen  
eltex-v1  pts/44   :pts/71:S.0    14:03      15:24   0.09s  33.75s SCREEN -S eltex  
eltex-v1  pts/51   :pts/76:S.0    14:28      29.00s  0.06s  0.10s SCREEN -S eltex  
eltex-v2  pts/55   :pts/23:S.0    Mon16      22:42m  1:42   7.19s SCREEN -S TOP -  
eltex-v1  pts/71   212.15.62.158  12:29      15:24   0.04s  0.01s screen -r eltex  
eltex-v1  pts/76   212.164.38.187 14:12      29.00s  0.02s  ?      screen -S eltex  
eltex-v3  pts/77   :pts/20:S.0    15:11      1.00s  0.05s  0.05s w  
eltex-v3  pts/78   176.208.33.209 13:03      2:01m  0.03s  0.03s -bash  
eltex-v1  pts/79   :pts/35:S.0    Mon13      25:46m  1:53   4.88s SCREEN -S top -  
eltex-v2  pts/89   193.17.95.152  13:33      4:14   0.06s  0.06s -bash  
eltex-v3  pts/99   176.208.33.209 13:37      1:23m  0.03s  0.03s -bash  
eltex-v2  pts/96   :pts/103:S.0   Mon12      26:25m  1.87s  1.87s top  
eltex-v37@eltex-2026-summer:~$
```

Рисунок 5 – Список подключенных пользователей

screen -S top -d -m top

screen -ls

```
eltex-v37@eltex-2026-summer: ~  
[detached from 2220689.pts-20.eltex-2026-summer]  
eltex-v37@eltex-2026-summer:~$ screen -S top -d -m top  
eltex-v37@eltex-2026-summer:~$ screen -ls  
There are screens on:  
2227696.top (07/07/2026 03:25:18 PM) (Detached)  
2220689.pts-20.eltex-2026-summer (07/07/2026 03:20:07 PM) (Detached)  
2205334.pts-20.eltex-2026-summer (07/07/2026 03:08:50 PM) (Detached)  
26431.pts-88.eltex-2026-summer (07/06/2026 12:51:31 PM) (Detached)  
4 Sockets in /run/screen/S-eltex-v37.  
eltex-v37@eltex-2026-summer:~$
```

Рисунок 6 – запуск фоновой сессии «top» и вывод списка сессий

screen -r top

```
eltex-v37@eltex-2026-summer: ~
top - 15:37:11 up 1 day, 12:19, 16 users, load average: 0.00, 0.01, 0.00
Tasks: 415 total, 1 running, 408 sleeping, 6 stopped, 0 zombie
%Cpu(s): 0.5 us, 0.5 sy, 0.0 ni, 98.9 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 3914.7 total, 2730.3 free, 885.9 used, 581.9 buff/cache
MiB Swap: 976.0 total, 976.0 free, 0.0 used. 3028.7 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
 57451 eltex-v+  20   0  11644   5420  3260  S   0.7   0.1   1:58.62 top
 58554 eltex-v+  20   0  11640   5328  3172  S   0.7   0.1   1:54.94 top
15694 eltex-v+  20   0  11640   5512  3360  S   0.3   0.1   2:01.18 top
24827 eltex-v+  20   0  11820   5448  3304  S   0.3   0.1   2:33.96 top
26575 eltex-v+  20   0  11644   5456  3304  S   0.3   0.1   2:05.92 top
55471 eltex-v+  20   0   9484   3752  1940  S   0.3   0.1   0:01.80 screen
57309 eltex-v+  20   0  11644   5480  3324  S   0.3   0.1   1:57.91 top
68904 eltex-v+  20   0   6848   3552  2420  S   0.3   0.1   0:33.35 watch
2195114 eltex-v+ 20   0  18012   7036  5164  S   0.3   0.2   0:00.52 sshd
    1 root      20   0 168672  13304  9216  S   0.0   0.3   0:03.21 systemd
    2 root      20   0     0     0     0  S   0.0   0.0   0:00.02 kthreadd
    3 root       0 -20     0     0     0  I   0.0   0.0   0:00.00 rcu_gp
    4 root       0 -20     0     0     0  I   0.0   0.0   0:00.00 rcu_par_gp
    5 root       0 -20     0     0     0  I   0.0   0.0   0:00.00 slub_flushwq
    6 root       0 -20     0     0     0  I   0.0   0.0   0:00.00 netns
    8 root       0 -20     0     0     0  I   0.0   0.0   0:00.00 kworker/0:0H-eve+
   10 root       0 -20     0     0     0  I   0.0   0.0   0:00.00 mm_percpu_wq
```

Рисунок 7 – Подсоединение к сессии «top»

watch /usr/bin/vmstat

```
eltex-v37@eltex-2026-summer: ~
Every 2.0s: /usr/bin/vmstat eltex-2026-summer: Tue Jul 7 15:40:35 2026

procs -----memory----- ---swap-- -----io----- -system-- -----cpu-----
 r b  swpd  free  buff  cache   si   so    bi   bo    in  cs us sy id wa st
 1  0      0 2779056 88612 507404    0    0     0    2   74 133  0  0 99  0  0
```

Рисунок 8 – Команда watch /usr/bin/vmstat

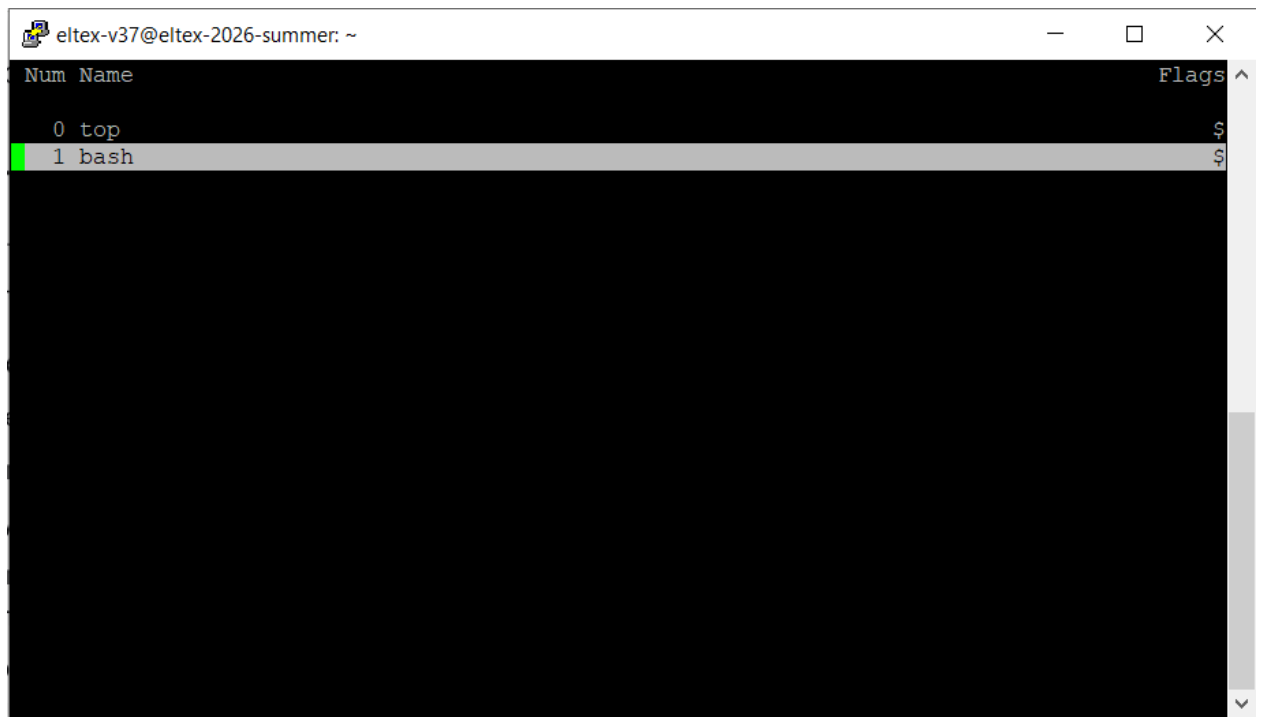


Рисунок 9 – Список окон

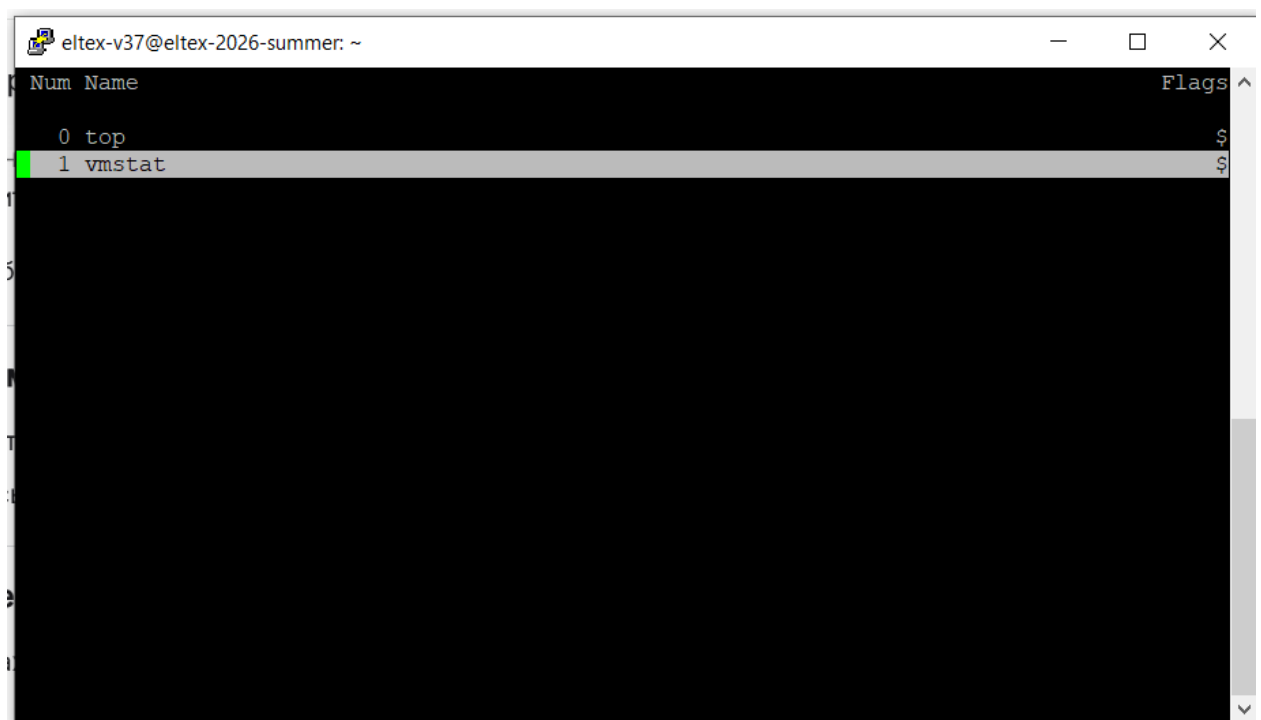


Рисунок 10 – Переименование второго окна

screen

vi report_part1.txt

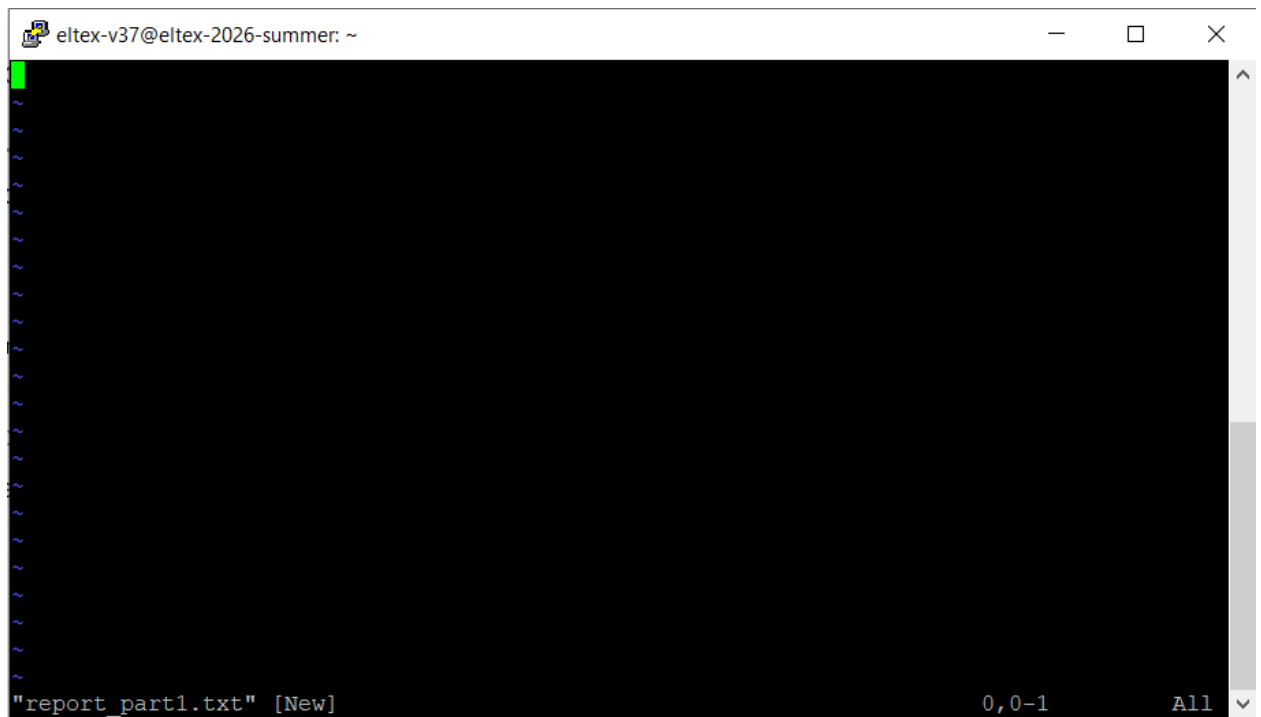


Рисунок 11 – Новая сессия и запуск команды vi report_part1.txt

screen -ls

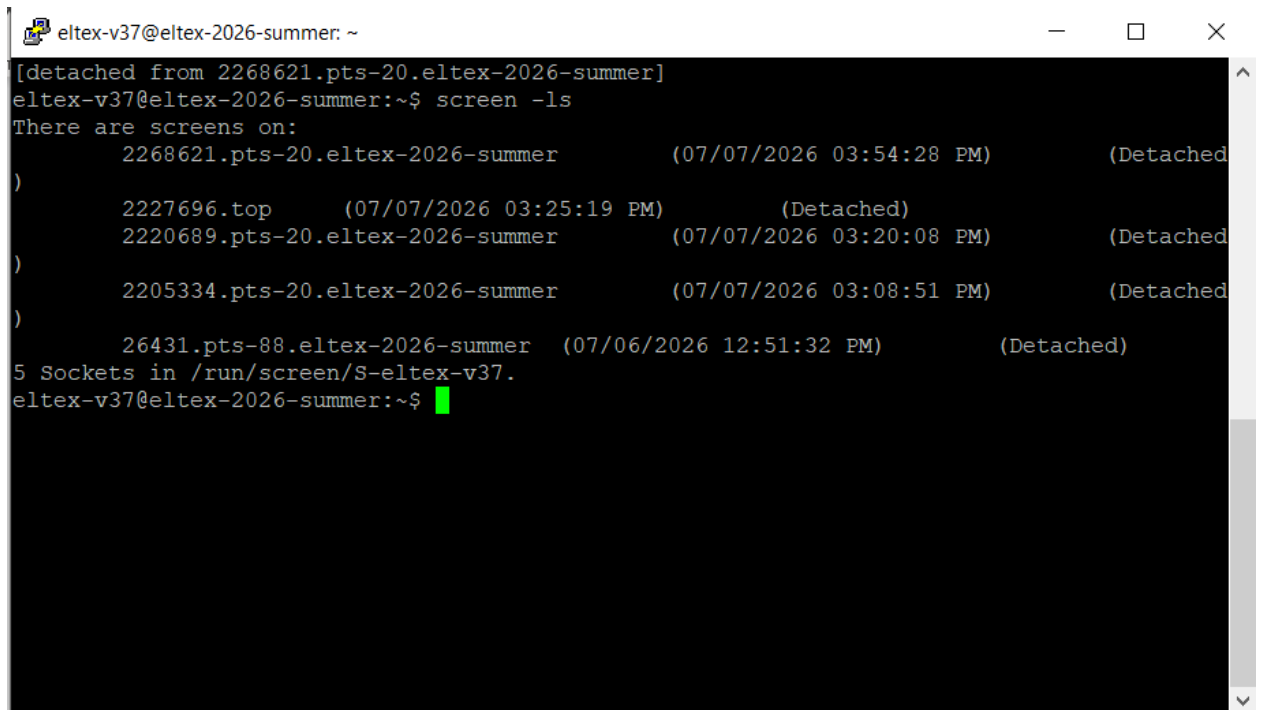


Рисунок 12 – Список сессий

logout

```

eltex-v37@eltex-2026-summer: ~
eltex-v37@217.71.138.1's password:
Linux eltex-2026-summer 6.1.0-50-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.176-1 (
2026-07-02) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Jul  7 15:11:06 2026 from 95.191.230.196
eltex-v37@eltex-2026-summer:~$ screen -ls
There are screens on:
    2268621.pts-20.eltex-2026-summer      (07/07/2026 03:54:28 PM)      (
Detached)
    2227696.top      (07/07/2026 03:25:19 PM)      (Detached)
    2220689.pts-20.eltex-2026-summer      (07/07/2026 03:20:08 PM)      (
Detached)
    2205334.pts-20.eltex-2026-summer      (07/07/2026 03:08:51 PM)      (
Detached)
    26431.pts-88.eltex-2026-summer      (07/06/2026 12:51:32 PM)      (Detache
d)
5 Sockets in /run/screen/S-eltex-v37.
eltex-v37@eltex-2026-summer:~$

```

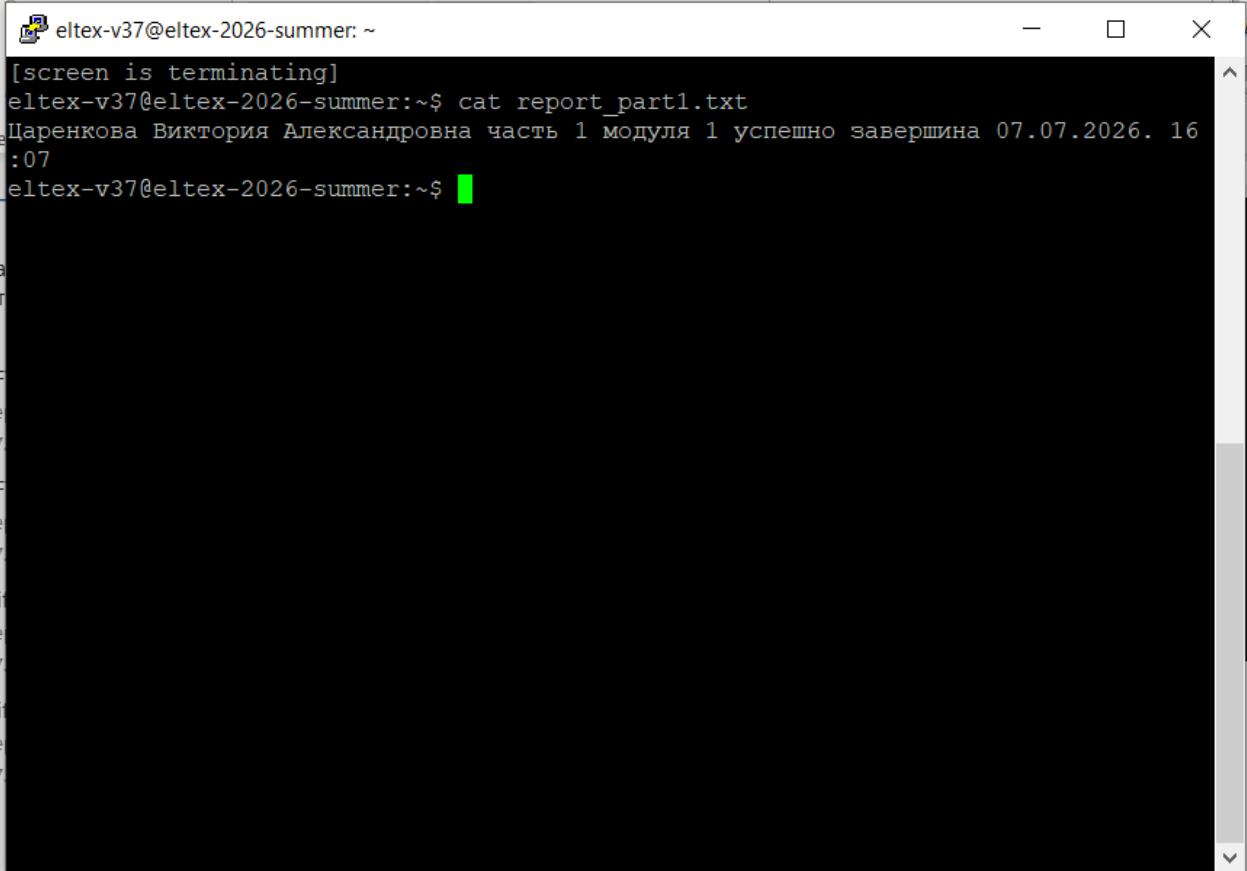
```
screen -r 2268621
```

A screenshot of a terminal window titled "eltex-v37@eltex-2026-summer: ~". The terminal displays a message in Russian: "Царенкова Виктория Александровна часть 1 модуля 1 успешно завершена 07.07.2026. 16:07". The cursor is positioned at the end of the second line. On the left side of the terminal, there are several blue tilde (~) symbols. At the bottom right, there is a status bar showing "1,142-85" and "All".

Рисунок 14 – Запись в файл

:wq

cat report_part1.txt

A terminal window titled 'eltex-v37@eltex-2026-summer: ~' with standard window controls. The terminal shows the command 'cat report_part1.txt' being executed. The output is displayed in Russian: 'Царенкова Виктория Александровна часть 1 модуля 1 успешно завершена 07.07.2026. 16:07'. The prompt 'eltex-v37@eltex-2026-summer:~\$' is followed by a green cursor. A message '[screen is terminating]' is visible at the top of the terminal output.

```
eltex-v37@eltex-2026-summer: ~  
[screen is terminating]  
eltex-v37@eltex-2026-summer:~$ cat report_part1.txt  
Царенкова Виктория Александровна часть 1 модуля 1 успешно завершена 07.07.2026. 16  
:07  
eltex-v37@eltex-2026-summer:~$
```

Рисунок 15 – Текст файла

screen -XS top quit

screen -XS 2220689 quit

screen -XS 2205334 quit

screen -XS 26431 quit

screen -ls

```
eltex-v37@eltex-2026-summer: ~  
[screen is terminating]  
eltex-v37@eltex-2026-summer:~$ cat report_part1.txt  
Царенкова Виктория Александровна часть 1 модуля 1 успешно завершена 07.07.2026. 16:07  
eltex-v37@eltex-2026-summer:~$ screen -ls  
There are screens on:  
2227696.top (07/07/2026 03:25:19 PM) (Detached)  
2220689.pts-20.eltex-2026-summer (07/07/2026 03:20:08 PM) (Detached)  
2205334.pts-20.eltex-2026-summer (07/07/2026 03:08:51 PM) (Detached)  
26431.pts-88.eltex-2026-summer (07/06/2026 12:51:32 PM) (Detached)  
4 Sockets in /run/screen/S-eltex-v37.  
eltex-v37@eltex-2026-summer:~$ screen -XS top quit  
eltex-v37@eltex-2026-summer:~$ screen -XS 2220689 quit  
eltex-v37@eltex-2026-summer:~$ screen -XS 2205334 quit  
eltex-v37@eltex-2026-summer:~$ screen -XS 26431 quit  
eltex-v37@eltex-2026-summer:~$ screen -ls  
No Sockets found in /run/screen/S-eltex-v37.  
eltex-v37@eltex-2026-summer:~$
```

Рисунок 16 – Закрытие всех сессий

Раздел 2

screen -S `whoami`_part2

script -q -c bash \${USER}_part.log

export TERM=dumb

ls

```
eltex-v37@eltex-2026-summer:~$ ls
ls          lsbl_release  lsinitramfs  lslocks      lsmod         lspci
lsattr      lscpu        lsipc        lslogins     lsns          lsusb
lsblk       lsfd         lsirq        lsmem        lsof
```

Рисунок 17 – Все команды, которые начинаются на «ls»

\$HIST

```
eltex-v37@eltex-2026-summer:~$ $HIST
$HISTCMD      $HISTCONTROL  $HISTFILE      $HISTFILESIZE  $HISTSIZ
eltex-v37@eltex-2026-summer:~$ $HIST
```

Рисунок 18 – Все переменные, которые начинаются с «HIST»

echo \$HISTFILESIZE

```
eltex-v37@eltex-2026-summer:~$ echo $HISTFILESIZE
2000
```

Рисунок 19 – Сколько команд может храниться в файле истории

unalias ls

ls -d ~/.*

```
eltex-v37@eltex-2026-summer:~$ ls -d ~/.*
/home/eltex-v37/.bash_history  /home/eltex-v37/.profile
/home/eltex-v37/.bash_logout  /home/eltex-v37/.ssh
/home/eltex-v37/.bashrc       /home/eltex-v37/.viminfo
/home/eltex-v37/.config       /home/eltex-v37/.Xauthority
eltex-v37@eltex-2026-summer:~$
```

Рисунок 20 – Имена файлов и каталогов из домашнего каталога, которые начинаются с «.»

export HISTTIMEFORMAT="%F %T "

history | tail -n 5

```

eltex-v37@eltex-2026-summer:~$ export HISTTIMEFORMAT="%F %T "
eltex-v37@eltex-2026-summer:~$ history | tail -n 5
  42  2026-07-07 18:38:30 echo $HISTFILESIZE
  43  2026-07-07 18:39:16 unalias ls
  44  2026-07-07 18:39:24 ls -d ~/.*
  45  2026-07-07 18:41:31 export HISTTIMEFORMAT="%F %T "
  46  2026-07-07 18:41:48 history | tail -n 5
eltex-v37@eltex-2026-summer:~$

```

Рисунок 21 – Настройка даты и времени

date

history | tail -n 2

tail -n 5 ~/.bash_history

export PROMPT_COMMAND="history -a"

whoami

tail -n 5 ~/.bash_history

history | tail -n 5

```

eltex-v37@eltex-2026-summer:~$ date
Tue Jul  7 06:44:04 PM +07 2026
eltex-v37@eltex-2026-summer:~$ history | tail -n 2
  47  2026-07-07 18:44:04 date
  48  2026-07-07 18:44:50 history | tail -n 2
eltex-v37@eltex-2026-summer:~$ tail -n 5 ~/.bash_history
export TERM=dumb
echo $HISTFILESIZE
ls -d ~/.*
\ls -d ~/.*
exit
eltex-v37@eltex-2026-summer:~$ export PROMPT_COMMAND="history -a"
eltex-v37@eltex-2026-summer:~$ whoami
eltex-v37
eltex-v37@eltex-2026-summer:~$ tail -n 5 ~/.bash_history
tail -n 5 ~/.bash_history
#1783424837
export PROMPT_COMMAND="history -a"
#1783424894
whoami
eltex-v37@eltex-2026-summer:~$ history | tail -n 5
  49  2026-07-07 18:45:44 tail -n 5 ~/.bash_history
  50  2026-07-07 18:47:17 export PROMPT_COMMAND="history -a"
  51  2026-07-07 18:48:14 whoami
  52  2026-07-07 18:48:30 tail -n 5 ~/.bash_history
  53  2026-07-07 18:49:19 history | tail -n 5
eltex-v37@eltex-2026-summer:~$

```

Рисунок 22 – Автоматическое сохранение набираемых команд в файле истории

```
DATE=$(date +%d.%m.%Y)
```

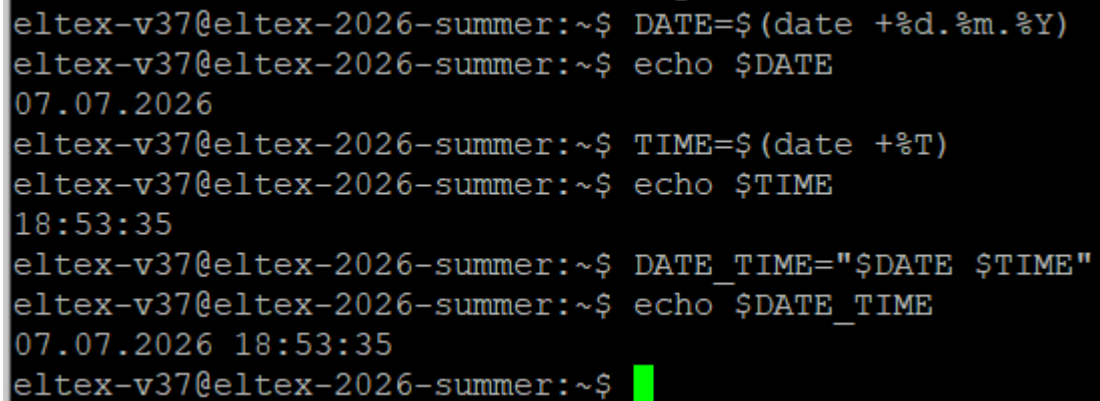
```
echo $DATE
```

```
TIME=$(date +%T)
```

```
echo $TIME
```

```
DATE_TIME="$DATE $TIME"
```

```
echo $DATE_TIME
```

A terminal window with a black background and white text. The prompt is 'eltex-v37@eltex-2026-summer:~\$'. The user enters 'DATE=\$(date +%d.%m.%Y)', followed by 'echo \$DATE' which outputs '07.07.2026'. Then 'TIME=\$(date +%T)' is entered, followed by 'echo \$TIME' which outputs '18:53:35'. Finally, 'DATE_TIME="\$DATE \$TIME"' is entered, followed by 'echo \$DATE_TIME' which outputs '07.07.2026 18:53:35'. The prompt is visible at the end of the last line.

```
eltex-v37@eltex-2026-summer:~$ DATE=$(date +%d.%m.%Y)
eltex-v37@eltex-2026-summer:~$ echo $DATE
07.07.2026
eltex-v37@eltex-2026-summer:~$ TIME=$(date +%T)
eltex-v37@eltex-2026-summer:~$ echo $TIME
18:53:35
eltex-v37@eltex-2026-summer:~$ DATE_TIME="$DATE $TIME"
eltex-v37@eltex-2026-summer:~$ echo $DATE_TIME
07.07.2026 18:53:35
eltex-v37@eltex-2026-summer:~$
```

Рисунок 23 – Текущие дата и время

```
ls /bin /sbin | grep '[0-9]'
```

```
eltex-v37@eltex-2026-summer: ~  
x86_64-linux-gnu-gcc-ranlib  
x86_64-linux-gnu-gcc-ranlib-12  
x86_64-linux-gnu-gcov  
x86_64-linux-gnu-gcov-12  
x86_64-linux-gnu-gcov-dump  
x86_64-linux-gnu-gcov-dump-12  
x86_64-linux-gnu-gcov-tool  
x86_64-linux-gnu-gcov-tool-12  
x86_64-linux-gnu-gold  
x86_64-linux-gnu-gp-archive  
x86_64-linux-gnu-gp-collect-app  
x86_64-linux-gnu-gp-display-html  
x86_64-linux-gnu-gp-display-src  
x86_64-linux-gnu-gp-display-text  
x86_64-linux-gnu-gprof  
x86_64-linux-gnu-gprofng  
x86_64-linux-gnu-ld  
x86_64-linux-gnu-ld.bfd  
x86_64-linux-gnu-ld.gold  
x86_64-linux-gnu-lto-dump  
x86_64-linux-gnu-lto-dump-12  
x86_64-linux-gnu-nm  
x86_64-linux-gnu-objcopy  
x86_64-linux-gnu-objdump  
x86_64-linux-gnu-ranlib  
x86_64-linux-gnu-readelf  
x86_64-linux-gnu-size  
x86_64-linux-gnu-strings  
x86_64-linux-gnu-strip  
dumpe2fs  
e2freefrag  
e2fsck  
e2image  
e2label  
e2mmpstatus  
e2scrub  
e2scrub_all  
e2undo  
e4crypt  
e4defrag  
fsck.ext2  
fsck.ext3  
fsck.ext4  
killall5  
mke2fs  
mkfs.ext2  
mkfs.ext3  
mkfs.ext4  
eltex-v37@eltex-2026-summer:~$
```

Рисунок 24 – Имена файлов, содержащие хотя бы одну цифру

```
PS1='$USER@$HOSTNAME-$(date "+%H:%M")>'  
export PS1  
echo "PS1='\u@\h-\$(date \"+%H:%M\")> ' >> ~/.bashrc
```

```
<8>echo "PS1='\u@\h-\$(date +%H:%M\)"> ' " >> ~/.bashrc
eltex-v37@eltex-2026-summer-19:11>bash
eltex-v37@eltex-2026-summer-19:11> exit
exit
eltex-v37@eltex-2026-summer-19:11>
```

Рисунок 25 – Вывод приглашения

```
mkdir -p ~/${date +%Y}-{01..12}
```

```
ls -d ~/${date +%Y}-*
```

```
eltex-v37@eltex-2026-summer-19:11>mkdir -p ~/${date +%Y}-{01..12}
eltex-v37@eltex-2026-summer-19:15>ls -d ~/${date +%Y}-*
/home/eltex-v37/2026-01 /home/eltex-v37/2026-05 /home/eltex-v37/2026-09
/home/eltex-v37/2026-02 /home/eltex-v37/2026-06 /home/eltex-v37/2026-10
/home/eltex-v37/2026-03 /home/eltex-v37/2026-07 /home/eltex-v37/2026-11
/home/eltex-v37/2026-04 /home/eltex-v37/2026-08 /home/eltex-v37/2026-12
eltex-v37@eltex-2026-summer-19:15>
```

Рисунок 26 – Подкаталоги для каждого месяца текущего года

Раздел 3

```
cd ~
```

```
echo "cd ~" >> $(whoami)_part3.log
```

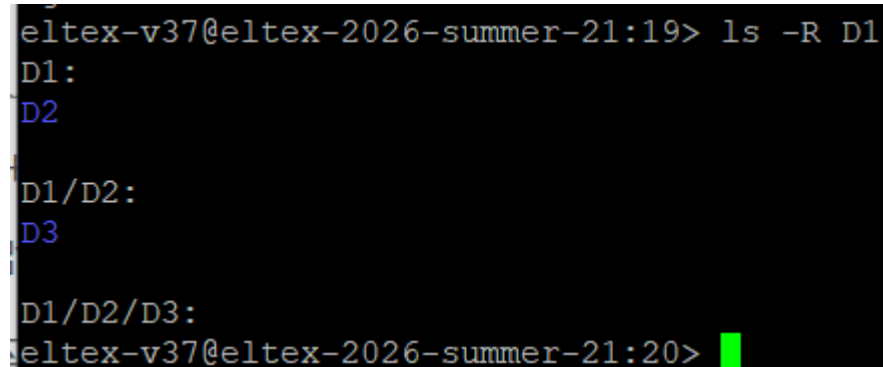
```
touch $(whoami)_part3.log
```

```
echo "touch \$(whoami)_part3.log" >> $(whoami)_part3.log
```

```
mkdir -p D1/D2/D3
```

```
echo "mkdir -p D1/D2/D3" >> $(whoami)_part3.log
```

```
ls -R D1
```



```
eltex-v37@eltex-2026-summer-21:19> ls -R D1
D1:
D2
D1/D2:
D3
D1/D2/D3:
eltex-v37@eltex-2026-summer-21:20>
```

Рисунок 27 – Проверка результата

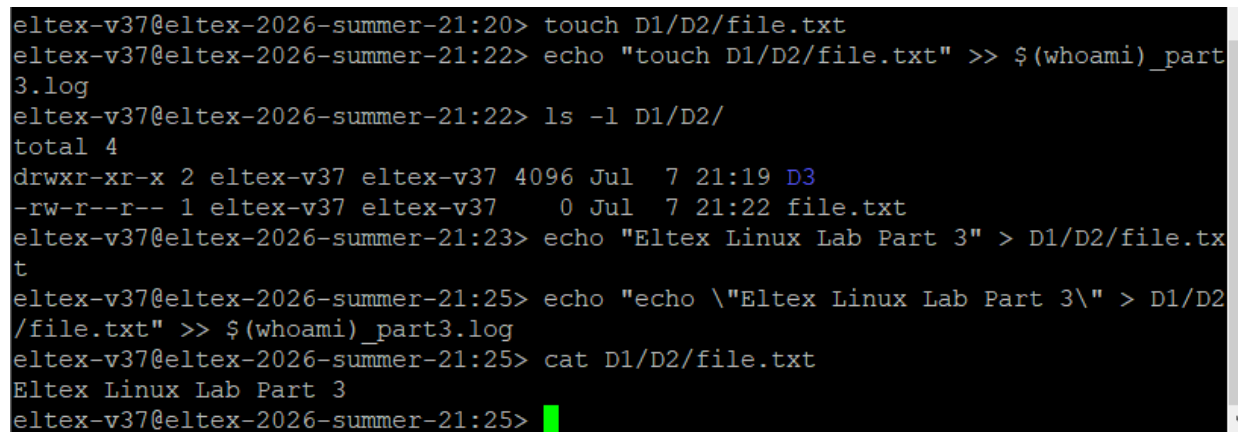
```
touch D1/D2/file.txt
```

```
echo "touch D1/D2/file.txt" >> $(whoami)_part3.log
```

```
echo "Eltex Linux Lab Part 3" > D1/D2/file.txt
```

```
echo "echo \"Eltex Linux Lab Part 3\" > D1/D2/file.txt" >> $(whoami)_part3.log
```

```
cat D1/D2/file.txt
```



```
eltex-v37@eltex-2026-summer-21:20> touch D1/D2/file.txt
eltex-v37@eltex-2026-summer-21:22> echo "touch D1/D2/file.txt" >> $(whoami)_part3.log
eltex-v37@eltex-2026-summer-21:22> ls -l D1/D2/
total 4
drwxr-xr-x 2 eltex-v37 eltex-v37 4096 Jul  7 21:19 D3
-rw-r--r-- 1 eltex-v37 eltex-v37    0 Jul  7 21:22 file.txt
eltex-v37@eltex-2026-summer-21:23> echo "Eltex Linux Lab Part 3" > D1/D2/file.txt
eltex-v37@eltex-2026-summer-21:25> echo "echo \"Eltex Linux Lab Part 3\" > D1/D2/file.txt" >> $(whoami)_part3.log
eltex-v37@eltex-2026-summer-21:25> cat D1/D2/file.txt
Eltex Linux Lab Part 3
eltex-v37@eltex-2026-summer-21:25>
```

Рисунок 28 – Запись текста в файл

```
ln D1/D2/file.txt D1/D2/D3/hardlink.txt
```

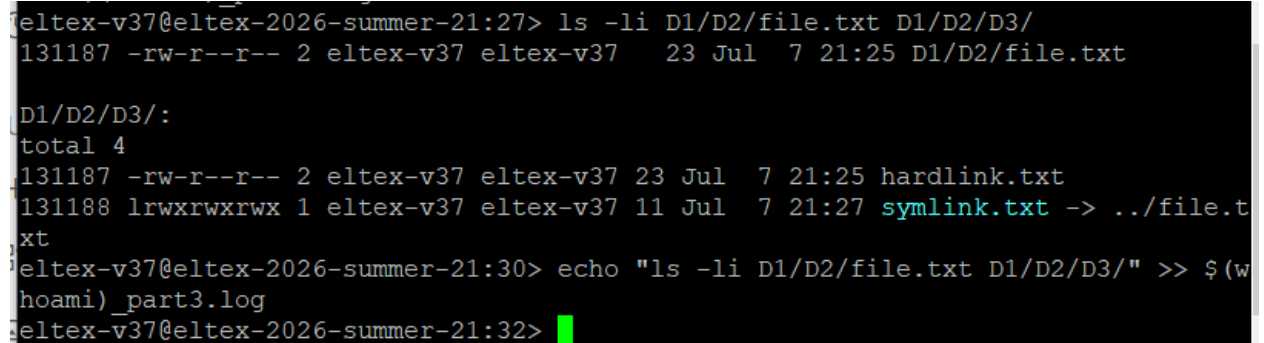
```
ln -s ../file.txt D1/D2/D3/symlink.txt
```

```
echo "ln D1/D2/file.txt D1/D2/D3/hardlink.txt" >> $(whoami)_part3.log
```

```
echo "ln -s ../file.txt D1/D2/D3/symlink.txt" >> $(whoami)_part3.log
```

```
ls -li D1/D2/file.txt D1/D2/D3/
```

```
echo "ls -li D1/D2/file.txt D1/D2/D3/" >> $(whoami)_part3.log
```



```
eltex-v37@eltex-2026-summer-21:27> ls -li D1/D2/file.txt D1/D2/D3/
131187 -rw-r--r-- 2 eltex-v37 eltex-v37 23 Jul 7 21:25 D1/D2/file.txt

D1/D2/D3/:
total 4
131187 -rw-r--r-- 2 eltex-v37 eltex-v37 23 Jul 7 21:25 hardlink.txt
131188 lrwxrwxrwx 1 eltex-v37 eltex-v37 11 Jul 7 21:27 symlink.txt -> ../file.t
xt
eltex-v37@eltex-2026-summer-21:30> echo "ls -li D1/D2/file.txt D1/D2/D3/" >> $(w
hoami)_part3.log
eltex-v37@eltex-2026-summer-21:32>
```

Рисунок 28 – Символическая и жесткая ссылки

```
mv D1/D2/file.txt D1/file.txt
```

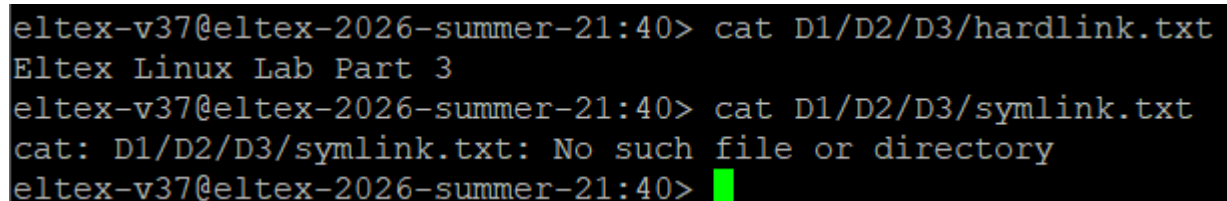
```
echo "mv D1/D2/file.txt D1/file.txt" >> $(whoami)_part3.log
```

```
cat D1/D2/D3/hardlink.txt
```

```
cat D1/D2/D3/symlink.txt
```

```
echo "cat D1/D2/D3/hardlink.txt" >> $(whoami)_part3.log
```

```
echo "cat D1/D2/D3/symlink.txt" >> $(whoami)_part3.log
```



```
eltex-v37@eltex-2026-summer-21:40> cat D1/D2/D3/hardlink.txt
Eltex Linux Lab Part 3
eltex-v37@eltex-2026-summer-21:40> cat D1/D2/D3/symlink.txt
cat: D1/D2/D3/symlink.txt: No such file or directory
eltex-v37@eltex-2026-summer-21:40>
```

Рисунок 29 – Проверка работы ссылок после перемещения файла

```
rm -rf D1/D2
```

```
echo "rm -rf D1/D2" >> $(whoami)_part3.log
```

```
find / -type f -size +20M -exec du -h {} + 2>/dev/null >
~/$$(whoami)_part3_gt20M.log
```

```
echo 'find / -type f -size +20M -exec du -h {} + 2>/dev/null >
~/$$(whoami)_part3_gt20M.log' >> $(whoami)_part3.log
```

```
find ~ -type f -mtime -1
```

echo 'find ~ -type f -mtime -1' >> \$(whoami)_part3.log

```
eltex-v37@eltex-2026-summer-21:53> find ~ -type f -mtime -1
/home/eltex-v37/.viminfo
/home/eltex-v37/.ssh/authorized_keys
/home/eltex-v37/report_part1.txt
/home/eltex-v37/eltex-v37_part.log
/home/eltex-v37/D1/file.txt
/home/eltex-v37/.bash_history
/home/eltex-v37/.Xauthority
/home/eltex-v37/.bashrc
/home/eltex-v37/eltex-v37_part3_gt20M.log
/home/eltex-v37/eltex-v37_part3.log
eltex-v37@eltex-2026-summer-22:01> echo 'find ~ -type f -mtime -1' >> $(whoami)_
part3.log
eltex-v37@eltex-2026-summer-22:01>
```

Рисунок 30 – Файлы, которые были изменены в течение последних 24х часов

which find

echo 'which find' >> \$(whoami)_part3.log

```
eltex-v37@eltex-2026-summer-22:01> which find
/usr/bin/find
eltex-v37@eltex-2026-summer-22:05> echo 'which find' >> $(whoami)_part3.log
eltex-v37@eltex-2026-summer-22:05>
```

Рисунок 31 – Команда find

file \$(which find)

echo 'file \$(which find)' >> \$(whoami)_part3.log

```
eltex-v37@eltex-2026-summer-22:05> file $(which find)
/usr/bin/find: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamica
lly linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=5545a6c7162d3
25f5e5ffabcbfe4e5521b66ce70, for GNU/Linux 3.2.0, stripped
eltex-v37@eltex-2026-summer-22:08> echo 'file $(which find)' >> $(whoami)_part3.
log
eltex-v37@eltex-2026-summer-22:08>
```

Рисунок 32 – Характер содержимого файла find

file /boot/initrd.img*

echo 'file /boot/initrd.img*' >> \$(whoami)_part3.log

```
eltex-v37@eltex-2026-summer-22:08> file /boot/initrd.img*
/boot/initrd.img-6.1.0-44-amd64: ASCII cpio archive (SVR4 with no CRC)
/boot/initrd.img-6.1.0-50-amd64: ASCII cpio archive (SVR4 with no CRC)
eltex-v37@eltex-2026-summer-22:18> echo 'file /boot/initrd.img*' >> $(whoami)_pa
rt3.log
eltex-v37@eltex-2026-summer-22:18>
```

Рисунок 33 – тип файлов /boot/initrd.img*